

Rui Zhang

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ACADEMIC EXPERIENCE

University of Pennsylvania , Master of Computer and Information Technology, GPA: 3.85	2021.01–2022.12
Coursework: Big Data Analytics, Database & Info System, Simulation Modeling & Analysis, Algorithms & Computation	
University of Shanghai for Science and Technology , <i>Bachelor of Prosthetic and Orthotic Engineering</i> , GPA: 3.71	2016.08–2020.05

Publications

- 8 peer-reviewed publications with **33+ citations** ([Google Scholar](#))
- Research topics include **machine learning**, **image processing**, and **data-driven modeling**

MACHINE LEARNING PROJECTS

LLM Retrieval and Evaluation Platform (RAG System)

- Built an end-to-end **RAG system** for question answering over external documents, including ingestion, chunking, embedding, retrieval, reranking, and LLM generation.
- Implemented **multi-stage retrieval stack** with dense vector search, **LLM-based query expansion (RAG-Fusion)**, **Reciprocal Rank Fusion (RRF)**, and **Cohere reranking** for improved recall and relevance ranking.
- Developed modular orchestration pipeline using LangChain to support **prompt templating**, **configurable retrieval strategies**, and **contextual compression**.
- Designed evaluation framework to benchmark retrieval relevance, answer quality, latency, and token cost across **chunk size**, **top-k retrieval**, **prompt variants**, and **reranking strategies**

WORK EXPERIENCE

Meta, Software Engineer (Instagram Ads, System, Product&ML)	2025.05–current
- Designed and implemented cross-stack latency optimization for the Instagram & Threads ads delivery pipeline, spanning AdFinder (C++), Django (Python), and WWW (PHP). Decoupled ad position serving from rendering within ML-driven feed ranking systems, reducing feed latency and enabling faster ad retrieval. - Led large-scale A/B experimentation and production rollout, achieving 1.6% lift in ad impressions and revenue uplift. - Diagnosed and resolved ranking logic bug in the Threads-in-Feed-Unit (TIFU) recommendation pipeline during Instagram → Threads inter-app navigation, restoring missing ad candidates and achieving 2% lift in Threads ad impressions. - Implemented dynamic Highest Position (HP) policy across AdFinder and Django layers to enable adaptive ad ranking strategies. Analyzed experimentation data from 100M+ users to guide ML-driven delivery optimization. - Leveraged LLM-based AI agents (Claude, Llama) to automate large-scale codebase refactoring and dead code detection, promoting AI tooling adoption and improving engineering productivity across the team.	
Microsoft, Software Engineer (Azure PostgreSQL, System&Storage)	2024.09–2025.5
- Developed Virtual Network (VNet) configuration validation for Azure PostgreSQL Flexible Server provisioning using C# and .NET Core, improving reliability for large-scale database workloads supporting analytics and ML pipelines. - Owned the ICM incident response queue, resolving complex cross-team infrastructure failures and implementing long-term mitigations such as automated replication slot cleanup, improving database cluster stability. - Collaborated across networking, storage, and database infrastructure teams to ensure high availability and fault tolerance for cloud-hosted PostgreSQL clusters.	
Amazon Web Service. Software Engineer (EC2 & EBS, System)	2022.05–2022.08(Internship), 2023.01–2024.09
- Designed a project aimed at identifying lost volumes within the EBS (Elastic Block Store) servers fleet, implemented multi-thread tasks & 2 SQS consumers to retrieve partitions metadata from the Redshift cluster and upload it to S3. - Developed a service that implements APIs for database (MySQL, QLDB) access and taking snapshots for EBS volumes, storing them in S3 buckets, also onboarded Canary as a monitoring system based on CloudWatch through Ruby. - Implemented EBS volume termination service for managing EBS snapshots/volumes given AES that vends account events like terminations/restores, integrated with SQS (DLQ) and DDB for event handling, ensuring GDPR compliance. - Built an auditing system to track fraud account termination by SQS and SNS, using Spring Boot to manage code components. - Designed an Integration Test (TestNG) based on the Regional Test on Demand server. Created code package, dependency set, environments in Apollo and new Pipeline from scratch, including infrastructure buildout and maintaining. - Developed an internal file sharing and management platform by using Java and Spring Boot for the backend, React.js and Redux for the frontend, and MongoDB for efficient data management, enabling seamless user access to information.	

SKILLS

Programming Languages: Python, C++, Java, SQL, JavaScript, etc.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), Vector Search, Embeddings, LLM Evaluation, Model Fine-Tuning

Data & AI Infrastructure: Spark, Kafka, Airflow, Docker, Kubernetes, AWS, GCP, Azure, Linux, MySQL, PostgreSQL, MongoDB, DynamoDB, Elasticsearch